

GIS-I: GEOGRAPHIC INFORMATION SYSTEMS

CHAPTER TWO GEOGRAPHIC DATA MODELS

BSc in Surveying Engineering

Chapter Learning Objectives

1. Explain the meaning of GIS data and geographic data sources.
2. Distinguish between spatial and non-spatial/attribute data.
3. Identify different sources of GIS data.
4. Explain the concept of a GIS data model.
5. Describe the vector spatial data model.
6. Explain point, line, and polygon representations.
7. Describe the raster spatial data model.
8. Explain raster cells, rows, columns, resolution, and raster values.
9. Compare vector and raster data models.
10. Select an appropriate data model for a particular GIS application.
11. Relate GIS data models to surveying, remote sensing, engineering, and environmental applications.

2.1 GIS Data and Data Sources

2.1.1 Introduction

Data is the foundation of every GIS project. A GIS may have powerful software and modern computers, but without appropriate and reliable data, it cannot produce useful information. In practical GIS work, a large part of the work involves finding data, collecting it, checking its quality, organizing it, and preparing it for analysis.

For example, suppose we want to prepare a map showing buildings, roads, rivers, and land parcels around a university campus. We need to know where these features are located and what characteristics they have. The answers are provided by GIS data.

GIS data is information about geographic phenomena that can be stored, managed, analyzed, and displayed using a GIS.

GIS data normally contains two closely related types of information:

- Spatial data
- Attribute data

2.1.2 Spatial Data

Spatial data describes the location, shape, size, and spatial relationships of geographic features. It answers the question: Where is it?

Examples include:

- Location of a building
- Location of a road
- Location of a river
- Location of a land parcel
- Location of a survey control point

For example, a GNSS receiver may record a water point at a particular Easting and Northing. Those coordinates provide the spatial location of the water point.

2.1.3 Attribute Data

Attribute data describes the characteristics or properties of a geographic feature. It answers the question: What is it?

ID	Type	Depth	Condition	Installation Year
W001	Borehole	85 m	Good	2018
W002	Well	25 m	Fair	2016
W003	Spring	—	Good	2020

The location of each water point is spatial data, while the information in the table is attribute data. A GIS becomes much more useful when spatial and attribute information are connected.

2.1.4 Spatial and Attribute Data Together

Consider a road layer. The spatial component represents the road's geometry as a line, while the attribute component may contain its name, surface type, width, and condition.

Road_ID	Road Name	Surface	Width	Condition
R001	Main Road	Asphalt	7 m	Good
R002	Campus Road	Gravel	5 m	Fair

The line shows where the road is, while the attribute table describes what the road is like. This relationship is one of the most fundamental concepts in GIS.

2.1.5 Types of GIS Data

For this course, the most important classification is:

- Spatial data – represents geographic location and geometry.
- Attribute data – represents descriptive information associated with geographic features.

Spatial data can further be represented using:

- Vector data: point, line, and polygon.
- Raster data: grid cells or pixels.

2.1.6 Sources of GIS Data

12. Field surveying
13. GNSS/GPS
14. Remote sensing
15. Aerial photography
16. UAV/drone surveys
17. Existing maps
18. Digitization
19. Government databases
20. Census data
21. Existing GIS databases
22. GPS-enabled mobile devices
23. Digital Elevation Models
24. Web and open geospatial datasets

2.1.7 Field Surveying

Surveying is one of the most important sources of accurate GIS data. Total stations, GNSS receivers, leveling instruments, and digital levels can collect accurate coordinates and elevations for buildings, roads, property boundaries, drainage structures, and control points.

Point	Easting	Northing	Elevation	Description
P001	500120.25	785430.12	1452.32	Building corner
P002	500135.18	785425.65	1452.18	Building corner
P003	500142.73	785440.26	1453.10	Road point

These coordinates can be imported into GIS and used to construct spatial features.

2.1.8 GNSS/GPS Data

Global Navigation Satellite Systems (GNSS) provide geographic coordinates using satellite-based positioning. Examples include GPS, GLONASS, Galileo, and BeiDou. GNSS data can be used for control points, roads, buildings, water points, utility poles, and boundary points.

2.1.9 Remote Sensing Data

Remote sensing provides information about the Earth's surface without direct physical contact with the observed object. Common sources include Landsat, Sentinel, MODIS, high-resolution satellite imagery, and aerial imagery.

- Land-cover mapping
- Vegetation analysis
- Agricultural monitoring
- Environmental monitoring
- Urban expansion
- Water-body mapping
- Change detection

2.1.10 Aerial Photography and UAV Data

Aerial photographs and UAV/drone imagery can provide very high-resolution information. Processed UAV data can produce orthophotos, digital surface models, point clouds, 3D models, and detailed maps.

2.1.11 Existing Maps

Existing topographic, cadastral, geological, administrative, road, and land-use maps can be important GIS data sources. Paper or scanned maps may need to be scanned, georeferenced, digitized, edited, and stored in a GIS database.

2.1.12 Census and Statistical Data

Census data can be linked to geographic units such as regions, zones, districts, kebeles, and enumeration areas.

Kebele	Population	Area
A	8,500	15 km ²
B	6,200	12 km ²
C	11,300	18 km ²

2.1.13 Data Quality

Before using a dataset, a GIS user should ask where it came from, when it was collected, what coordinate system was used, and whether its accuracy is suitable for the intended purpose.

- Positional accuracy – how close the recorded location is to the true location.
- Attribute accuracy – how correct the descriptive values are.
- Completeness – whether all required features are included.
- Logical consistency – whether the data structure and relationships are correct.
- Temporal accuracy – whether the data represents the required time period.

2.1.14 Primary and Secondary Data

Primary data is collected directly for the intended project, whereas secondary data has already been collected and is reused.

Primary Data	Secondary Data
GNSS survey	Existing topographic map
Total station survey	Government GIS dataset
UAV survey	Satellite imagery
Field observations	Census data

2.2 GIS Data Model

2.2.1 What is a Data Model?

A data model is a conceptual way of representing real-world phenomena in a computer. The real world is complex, so GIS uses simplified digital representations that computers can store and analyze.

For example, a real road has width, length, curves, surface, elevation, intersections, traffic, drainage, and signs. In GIS it may be simplified to a line feature, while its characteristics are stored in an attribute table.

A GIS data model is a simplified representation of real-world geographic phenomena designed for computer storage, management, analysis, and visualization.

2.2.2 Why Do We Need Data Models?

- Represent geographic location.
- Represent shape and spatial relationships.
- Store attributes.
- Perform analysis.
- Update information.
- Visualize geographic information.

2.2.3 Major GIS Spatial Data Models

- Vector Data Model – represents geographic features using points, lines, and polygons.
- Raster Data Model – represents geographic space as rows, columns, and cells.

2.2.4 Object-Based and Field-Based Representation

The object view considers the world as a collection of identifiable objects, such as buildings, roads, parcels, and bridges. The field view considers the world as continuously varying phenomena, such as elevation, temperature, rainfall, and soil moisture.

View	Typical Examples	Common Representation
Object view	Building, road, parcel, bridge	Vector
Field view	Elevation, rainfall, temperature	Raster

2.3 Vector Spatial Data Model

2.3.1 Introduction

The vector data model represents geographic features using geometric objects. The three fundamental vector geometries are point, line, and polygon.

2.3.2 Point Data

A point represents a geographic feature whose location is important but whose size is not significant at the map scale.

- Wells
- Survey control points
- Trees
- Electric poles
- Schools
- Hospitals
- Weather stations
- Landmarks

ID	Easting	Northing	Feature
P001	500250	785430	Water point
P002	500450	785620	Control point
P003	500710	785850	Electric pole

2.3.3 Line Data

A line represents a geographic feature with length and is constructed from a sequence of coordinates called vertices.

- Roads
- Rivers
- Pipelines
- Power lines
- Railway lines
- Drainage channels

2.3.4 Polygon Data

A polygon represents a geographic feature with an enclosed area. It is formed by a closed sequence of line segments.

- Land parcels
- Buildings
- Lakes
- Administrative areas
- Forests
- Agricultural fields
- Land-use zones

2.3.5 Vector Representation of Common Features

Real-world Feature	Vector Geometry
Survey control point	Point
Water point	Point
Electric pole	Point
Road	Line
River	Line
Pipeline	Line
Land parcel	Polygon
Building footprint	Polygon
Lake	Polygon
Administrative boundary	Polygon

2.3.6 Coordinates and Vertices

Vector features are constructed from coordinates. A line is formed by connecting vertices in sequence. For polygons, the vertices are connected to form a closed boundary.

Vertex	X	Y
1	500000	785000
2	500120	785080
3	500250	785210
4	500420	785350

2.3.7 Vector Attributes

Road_ID	Type	Surface	Width	Length
R001	Main	Asphalt	7 m	2.5 km
R002	Local	Gravel	5 m	1.2 km
R003	Access	Earth	4 m	0.8 km

2.3.8 Advantages of Vector Data

- Accurately represents discrete features.
- Provides precise coordinate-based geometry.
- Often stores discrete features efficiently.
- Supports detailed feature attributes.

- Well suited to network analysis.

2.3.9 Limitations of Vector Data

- Geometric relationships can be complex to manage.
- Complex polygon overlay may require substantial processing.
- Less natural for continuously varying phenomena such as temperature and elevation.

2.3.10 Vector Data and Topology

Topology describes important spatial relationships between geographic features, including connectivity, adjacency, containment, and intersection.

Topology is especially important in cadastral mapping, transportation networks, utility networks, and administrative boundary datasets.

2.3.11 Common Vector Data Formats

- Shapefile
- GeoPackage
- GeoJSON
- File Geodatabase feature classes
- KML/KMZ

A traditional Shapefile normally works as a group of related files. Common components include .shp for geometry, .shx for shape indexing, .dbf for attributes, and .prj for coordinate reference system information.

2.4 Raster Spatial Data Model

2.4.1 Introduction

The raster data model represents geographic space as a regular grid of cells. Each cell stores a value. A raster can therefore be viewed as a large matrix of values arranged in rows and columns.

12	15	18	20
10	14	17	22
8	12	16	21
7	10	14	19

2.4.2 Raster Cell

The basic unit of a raster dataset is called a cell or pixel. Each cell represents an area on the Earth's surface.

For example, a 10 m × 10 m raster cell represents 100 m² on the ground, while a 30 m × 30 m cell represents 900 m².

2.4.3 Raster Resolution

Spatial resolution refers to the ground dimensions represented by each raster cell. Common examples are 1 m, 10 m, 30 m, and 100 m.

Generally, smaller cells provide greater spatial detail but require more storage and processing.

2.4.4 Raster Rows and Columns

A raster consists of rows, columns, and cells. A raster containing 1,000 rows and 1,000 columns contains 1,000,000 cells.

2.4.5 Raster Values

The value stored in a raster cell depends on what the raster represents.

- Elevation raster: a cell may contain 1452 m.
- Temperature raster: a cell may contain 24.5°C.
- Land-cover raster: a cell may contain a class code such as 1 = forest or 2 = agriculture.
- Satellite imagery: a cell may contain a digital value recorded by a sensor.

2.4.6 Continuous and Categorical Raster Data

Continuous data varies continuously across space, while categorical data represents discrete classes.

Raster Type	Examples
Continuous	Elevation, temperature, rainfall, soil moisture
Categorical	Land use, land cover, soil type, geology

2.4.7 Examples of Raster Data

- Satellite imagery
- Aerial imagery
- Digital Elevation Models
- Land-cover maps
- Rainfall surfaces
- Temperature surfaces
- Slope maps
- Aspect maps
- Vegetation-index maps such as NDVI

2.4.8 Digital Elevation Model (DEM)

A Digital Elevation Model is a raster dataset representing elevation values across an area. DEMs are particularly important for Surveying Engineering.

Cell	Elevation
A1	673 m
A2	680 m
A3	691 m
B1	720 m
B2	735 m
B3	748 m

From a DEM, GIS can derive:

- Contour lines
- Slope
- Aspect

- Hillshade
- Watersheds
- Drainage networks
- Flow direction
- Flow accumulation
- Terrain profiles

2.4.9 Raster and Satellite Imagery

Satellite imagery is naturally represented using raster structures. Each spectral band is normally a raster containing pixel values. GIS and remote sensing software can combine bands to calculate indices such as NDVI.

$$\text{NDVI} = (\text{NIR} - \text{Red}) / (\text{NIR} + \text{Red})$$

The resulting NDVI surface is also represented as a raster.

2.4.10 Advantages of Raster Data

- Simple grid structure.
- Excellent for continuous phenomena.
- Efficient for mathematical and cell-by-cell analysis.
- Natural representation for remote sensing imagery.
- Very suitable for terrain analysis.

2.4.11 Limitations of Raster Data

- High-resolution rasters can require large storage.
- Spatial detail depends strongly on cell size.
- Boundaries may appear stepped or pixelated.
- A simple raster band normally stores one principal value per cell, so feature-level attribute structures are less flexible than in vector data.

2.4.12 Vector versus Raster Data Model

Characteristic	Vector	Raster
Basic unit	Point, line, polygon	Cell/pixel
Representation	Discrete objects	Grid surface
Coordinates	Explicit geometric coordinates	Cell position
Best for	Discrete features	Continuous surfaces
Examples	Roads, parcels, buildings	DEM, satellite imagery
Boundaries	Usually precise	Dependent on resolution
Attribute structure	Feature-based	Cell/value-based
Network analysis	Very suitable	Less suitable
Terrain analysis	Possible	Very suitable
Remote sensing	Can represent derived features	Natural representation
Storage	Often efficient for discrete features	Can become large
Spatial resolution	Coordinate-based	Cell-size based

2.4.13 Example: Representing a River

In a vector model, a river can be represented as a line with attributes such as river name, length, stream order, and water quality. In a raster model, the river can be represented by a group of cells classified as river. The appropriate model depends on the purpose of the analysis.

2.4.14 Example: Representing Elevation

Elevation is generally a continuous phenomenon. A raster DEM can represent elevation at every cell and can be used to calculate slope, aspect, contours, hillshade, and drainage.

673	681	690
700	715	728
740	755	770

2.4.15 Example: Land Parcels

Land parcels are discrete features with clearly defined boundaries. A vector polygon is generally appropriate. Each parcel can have attributes such as parcel identification, area, land use, and ownership type.

Parcel_ID	Area	Land Use	Ownership Type
P001	450 m ²	Residential	Private
P002	650 m ²	Commercial	Private
P003	1,200 m ²	Institutional	Public

2.4.16 Choosing Between Vector and Raster

The choice should depend on the nature of the geographic phenomenon and the intended analysis.

Use Vector When	Use Raster When
Features are discrete	Phenomenon varies continuously
Boundaries are important	Data originates from satellite imagery
Individual objects need individual attributes	Cell-by-cell analysis is required
Network analysis is required	Terrain analysis is required
Cadastral accuracy is important	Suitability modeling is required

2.4.17 Vector-Raster Integration

In practical GIS projects, vector and raster data are often used together. For example, a site-selection project may combine vector layers such as roads, rivers, settlements, and administrative boundaries with raster layers such as elevation, slope, rainfall, and land cover.

2.4.18 Example: Selecting a Suitable Site for a Health Center

Suppose a health center should be close to settlements and roads, avoid flood-prone areas and steep slopes, and remain outside environmentally sensitive areas.

- Vector: settlements, roads, rivers, administrative boundaries.
- Raster: slope, elevation, land cover, flood-risk surface.

The datasets can be reclassified, buffered, overlaid, and combined to produce a suitability map. This demonstrates that vector and raster are complementary rather than competing data models.

Chapter Summary

GIS requires data to represent the real world in a digital environment. GIS data can be broadly divided into spatial data and attribute data. Spatial data describes where geographic phenomena occur, while attribute data describes their characteristics.

A GIS data model provides a structured way of representing real-world geographic phenomena in a computer. The two fundamental spatial data models are vector and raster.

The vector data model represents discrete geographic objects using points, lines, and polygons. It is particularly appropriate for survey points, roads, buildings, land parcels, administrative boundaries, and utility networks.

The raster data model represents geographic space as a regular grid of cells. Each cell stores a value. Raster data is particularly appropriate for continuous phenomena such as elevation, temperature, rainfall, and soil moisture, as well as satellite imagery.

Neither model is universally better than the other. The appropriate model depends on the nature of the phenomenon, the purpose of the GIS project, required accuracy, type of analysis, and available data. In practical projects, vector and raster data are frequently integrated.

Review Questions

A. Short-Answer Questions

25. What is GIS data?
26. What is spatial data?
27. What is attribute data?
28. Differentiate between spatial and attribute data.
29. List five sources of GIS data.
30. What is primary GIS data?
31. What is secondary GIS data?
32. What is a GIS data model?
33. Why is a data model necessary in GIS?
34. Name the two major spatial data models.
35. What are the three basic vector geometries?
36. What is a raster cell?
37. What is raster resolution?
38. What is a Digital Elevation Model?
39. What is topology?
40. Give three examples of vector data.
41. Give three examples of raster data.
42. What is the difference between continuous and categorical raster data?

B. Discussion Questions

43. Explain the relationship between spatial and attribute data.
44. Discuss the major sources of GIS data.
45. Explain how surveying contributes to GIS data acquisition.
46. Explain the role of GNSS in GIS.
47. Discuss the major characteristics of the vector data model.

48. Discuss the major characteristics of the raster data model.
49. Compare the advantages and limitations of vector and raster data.
50. Explain why land parcels are normally represented as vector polygons.
51. Explain why DEMs are commonly represented as raster data.
52. Explain why vector and raster data are often integrated in GIS projects.

Practical Exercise 2.1: Identifying GIS Data

Students should identify at least 15 geographic features in their university or surrounding community. For each feature, determine its spatial representation, possible attributes, and data collection method.

No.	Geographic Feature	Spatial Representation	Possible Attributes	Data Collection Method
1	Building	Polygon	Name, type, floors	GNSS/digitizing
2	Road	Line	Surface, width	Survey/GIS
3	Water point	Point	Type, condition	GNSS/field survey
4	Parcel	Polygon	Area, land use	Cadastral survey
5	River	Line	Name, width	GPS/imagery
6	DEM	Raster	Elevation	Remote sensing
7	Tree	Point	Species, height	Field survey
8	Land cover	Raster	Class	Satellite imagery

Practical Exercise 2.2: Vector or Raster?

For each feature below, determine whether vector or raster would be more appropriate and provide a short justification.

Feature	Vector/Raster	Reason
Land parcel		
Road		
Building		
Elevation		
Rainfall		
Satellite image		
River network		
Temperature		
School location		
Land-cover map		

Practical Exercise 2.3: Understanding Raster Resolution

Consider a study area of 1 km × 1 km. Calculate the approximate number of cells required for 10 m, 30 m, and 100 m resolution.

Resolution	Cells Along One Side	Approximate Total Cells
10 m	$1,000 / 10 = 100$	$100 \times 100 = 10,000$
30 m	$1,000 / 30 \approx 33$	$33 \times 33 \approx 1,089$
100 m	$1,000 / 100 = 10$	$10 \times 10 = 100$

Discussion: Which raster provides the greatest spatial detail? Why does higher spatial resolution generally require more storage?

Practical Exercise 2.4: GIS Data Sources

Select a study area around your university. Identify suitable data sources for roads, buildings, rivers, administrative boundaries, elevation, land cover, population, and survey control points.

For each dataset, identify: data source, data type, spatial representation, coordinate system, expected accuracy, and possible use.

Assignment

Identification and Evaluation of GIS Data Sources for a Local Study Area

Select an area of approximately 5–10 km² around your university or local community. Identify at least 10 GIS datasets that could be used to describe the study area.

53. Describe what each dataset represents.
54. Identify its possible source.
55. State whether it is primary or secondary data.
56. State whether it is vector or raster.
57. Identify possible attributes.
58. Discuss expected accuracy.
59. Explain how the dataset could be used in GIS analysis.

Submission should include a short written report, a data-source table, at least one GIS map, and a brief discussion of data quality.

Key Terms

Attribute Data: Descriptive information associated with a geographic feature.

Cell/Pixel: The basic spatial unit of a raster dataset.

Continuous Data: Geographic information that varies continuously across space.

Data Model: A conceptual framework for representing real-world phenomena in digital form.

DEM: A raster representation of elevation values across geographic space.

Digitization: Conversion of geographic information into digital GIS features.

GNSS: Satellite-based positioning technology used to determine geographic coordinates.

Polygon: A closed vector geometry representing an area.

Point: A vector geometry representing a location.

Raster: A spatial data model consisting of a regular grid of cells.

Resolution: The ground size represented by one raster cell.

Spatial Data: Information describing the geographic location and geometry of features.

Vector: A spatial data model based on points, lines, and polygons.

Vertex: A coordinate used to construct a vector line or polygon.

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Recommended Reading for Students

For students encountering GIS for the first time, begin with Sutton, Dassau and Sutton for accessible concepts and practical GIS work. Use Chang for systematic treatment of GIS data, models, analysis, and applications; Burrough and McDonnell for deeper theoretical foundations; and Aronoff for understanding GIS as an information-management and decision-support system.